

Sutex Bank College of Computer Applications & Science, Amroli (BCA)

S.Y.B.C.A. (Sem-III)

Assignment-3 & 4

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Q.1 MCQ Questions

- 1. What is Polymorphism?**
 - A) Data hiding
 - B) One interface, many forms
 - C) Inheritance
 - D) Encapsulation
- 2. Compile-time polymorphism is achieved by:**
 - A) Virtual function
 - B) Function Overloading
 - C) Inheritance
 - D) Friend function
- 3. Run-time polymorphism is achieved by:**
 - A) Function Overloading
 - B) Operator Overloading
 - C) Function Overriding
 - D) Constructor
- 4. Which keyword is used for runtime polymorphism?**
 - A) static
 - B) virtual
 - C) friend
 - D) const
- 5. Function overloading means:**
 - A) Same function name with different parameters
 - B) Different function names
 - C) Same class only
 - D) Same return type only

6. Function overriding requires:

- A) Same class
- B) Different classes with inheritance
- C) Constructor
- D) Friend function

7. Friend function can access:

- A) Public members only
- B) Private members
- C) Protected members only
- D) None

8. Which function is called according to object type at runtime?

- A) Static function
- B) Inline function
- C) Virtual function
- D) Friend function

9. A pure virtual function is declared by:

- A) virtual void fun();
- B) virtual void fun()=0;
- C) void fun()=0;
- D) friend void fun();

10. Which is not polymorphism?

- A) Overloading
- B) Overriding
- C) Inheritance
- D) Virtual function

11. Which keyword prevents overriding?

- A) final
- B) virtual
- C) friend
- D) static

12.Data Structure is:

- A) Programming language
- B) Way of organizing data
- C) Database
- D) Operating System

13.Which is a linear data structure?

- A) Tree
- B) Graph
- C) Stack
- D) Heap

14.Which is a non-linear data structure?

- A) Queue
- B) Array
- C) Tree
- D) Stack

15.Stack follows:

- A) FIFO
- B) LIFO
- C) Random
- D) Circular

16.Queue follows:

- A) LIFO
- B) FIFO
- C) Random
- D) None

17.Stack overflow occurs when:

- A) Stack is empty
- B) Stack is full
- C) Stack has one element
- D) None

18.Stack underflow occurs when:

- A) Stack is full
- B) Stack is empty
- C) Stack has many elements
- D) None

19.Recursion means:

- A) Loop
- B) Function calling itself
- C) Class
- D) Object

20.Stack deletion takes place at:

- A) Front
- B) Rear
- C) Top
- D) Bottom

Q-2 Answer the following questions:

1. Explain Multiple Inheritance.
2. What is Constructor? Explain default of constructors.
3. Explain Characteristics of a Friend function.
4. What is stack? Write an algorithm to perform POP.
5. Define: Data Structure. List various data structures.